



Natural language processing - Submission 3: Final

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Abstract

General purpose large language models often lack reliable knowledge of specific legal systems and may hallucinate unsupported legal claims. In this project, we build a retrieval augmented generation (RAG) system for Slovenian legal question answering over real-estate and property-law legislation from the PISRS dataset. Legal texts are split into article level chunks, embedded with BGE-M3, stored in a FAISS index, and answered by GAMS3-12B-Instruct. We evaluate retrieval and answer quality on a 30 question development set and a 50 question article derived set, testing three GaMS model sizes, three embedding models, BM25 and hybrid retrieval, MMR, re-ranking, query expansion and generation parameter sweeps.

1. Introduction

Legal question answering is a knowledge intensive task, answers must not only sound plausible, but must be supported by the correct legal provisions. This is especially important in the legal domain, where unsupported or outdated claims can be misleading. Our goal was to build a Slovenian legal assistant that answers questions about real-estate and property-law legislation. The implemented system retrieves relevant legal sources and generates short answers.

We focus on retrieval augmented generation (RAG), where a language model is not expected to answer purely from its internal parameters. Instead, a retriever first selects relevant legal text passages and the generator answers based on those passages. This follows the general RAG idea introduced by Lewis et al. [1]. We use FAISS for vector similarity search [2], BGE-M3 as the main multilingual embedding model [3] and Slovenian GaMS models [4] as the generator. We also test multilingual-E5-large [5] and Slovlo-v1 [6].

2. Methods

2.1 Corpus and chunking

We used the PISRS legal documents from the COLESLAW PISRS data, especially `register-predpisov.jsonl`. We filtered the corpus with keywords related to real-estate such as *nepremičnine*, *zemljiška knjiga*, *kataster*, *služnost*, *hipoteka*, *etažna lastnina*, *gradbeno dovoljenje*, and *urejanje prostora*. Documents were split into article level chunks using article headings such as *170. člen*. The chunking script also

normalized Unicode, whitespace, punctuation, warning lines, and structural heading artifacts. Our final data and index statistics can be seen in Table 1.

Table 1. Final data and index statistics.

Item	Value
PISRS records read	7,375
Filtered legal documents kept	3,468
Final article chunks	78,904
Limited early chunks	23,592
Final FAISS index size	309 MB
Final metadata size	157 MB
Development evaluation set	30 questions
Final article-derived set	50 questions

2.2 RAG pipeline and models

The final pipeline has four steps. First, the user question is embedded. Second, FAISS retrieves the most similar legal article chunks. Third, the top contexts are inserted into a legal QA prompt and fourth, the GaMS model generates a short answer grounded in the retrieved text.

The final selected configuration was:

- Generator: `cjvt/GaMS3-12B-Instruct`
- Embedding: `BAAI/bge-m3`
- Retrieval: dense FAISS retrieval
- Reranker: not used in final setup
- MMR: not used in final setup

- Prompt: short extractive prompt P2
- Context chunks: `context_n = 2`
- Context length: 1200 characters per chunk
- Generation budget: 220 new tokens

3. Evaluation Setup

We used two main evaluation sets. The 30 question development set contained realistic user style questions: easy, hard, true/false, comparison, scenario, and tricky negative questions. It was useful for development, but some gold sources were ambiguous. The final dataset was generated with LLM assistance from randomly sampled PISRS article chunks, which we fed to ChatGPT Plus. For each sampled chunk, the LLM generated a realistic legal question and an expected answer hint based on the provided article text. Because the source chunk was known at generation time, we could treat it as the gold article for retrieval evaluation. This reduced ambiguity compared with manually labeling realistic user questions, where several different sources may be valid. The final 50 question article derived set contained 40 single source and 10 multi source questions. The question types are defined as seen in Table 2.

Table 2. Question difficulty and type categories.

Category	Description
Easy	Direct factual or definitional questions, usually answerable from one article.
Hard	Single source questions requiring careful reading, conditions, exceptions, or legal procedure.
Comparison	Questions comparing two concepts, records, or procedures.
Scenario	Practical situations asking which rule applies.
Yes/No	Questions requiring a yes/no answer and justification.
Tricky negative	Questions with false assumptions, negation, or overgeneralization.
Multi source	Questions requiring evidence from two or more articles.

Retrieval was evaluated with Top-1, Top-3, Top-5, and Top-10 accuracy/recall against the gold article(s). For multi source questions, we measured both whether at least one required source was retrieved and whether all required sources were retrieved. Answer quality was evaluated as *correct*, *partial*, or *wrong*. Answer evaluation was done manually and using LLM as a judge using the expected (gold) source articles, retrieved chunks, answer hint and generated answer.

4. Experiments and Results

Altogether we tested limited vs. full corpus indexing, improved chunk cleanup, three GaMS model sizes, no-RAG vs. RAG, five prompt variants, output cleaning, different context/generation parameter sweep, BGE reranking, MMR, query expansion, three embedding models, BM25 and hybrid BM25 with dense retrieval, multi source context size and general/unanswerable questions.

4.1 RAG, model size, and prompts

We compared no-RAG and RAG versions of GaMS-2B, GaMS-9B, and GaMS3-12B on the 30 question development set. RAG improved all model sizes substantially (Figure 1). Across all three models, no-RAG achieved 23/90 correct, 41/90 partial, and 26/90 wrong answers, while RAG achieved 63/90 correct, 20/90 partial, and only 7/90 wrong answers. This confirms that grounding the model in retrieved PISRS sources is more important than relying on parametric knowledge alone.

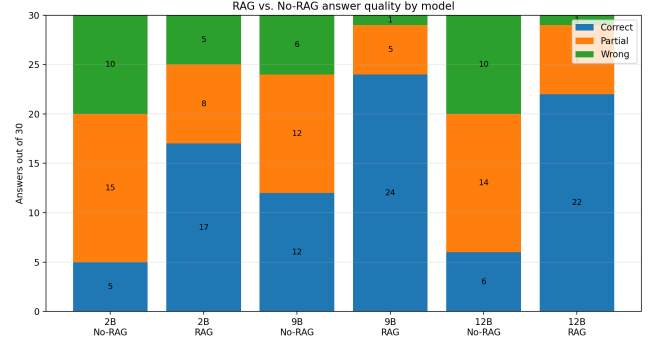


Figure 1. Comparison of no-RAG vs RAG for GaMS 2B, 9B and 12B models on the 30-question development set.

The best result on this set was GaMS-9B with RAG (24/30 correct), though we selected GaMS3-12B for final experiments because it produced cleaner and more complete answers in qualitative runs. We also tested five prompt variants and selected P2, a short extractive prompt, because it produced the fewest formatting artifacts. We added output cleanup: special token skipping, stop marker cleanup, markdown cleanup, and invalid citation removal. After cleanup, remaining errors were mostly legal correctness and source completeness, not formatting.

4.2 Embedding and retrieval experiments

Figure 2 shows the embedding comparison on the final 50-question set. BGE-M3 gave the best recall and was kept as the dense baseline.

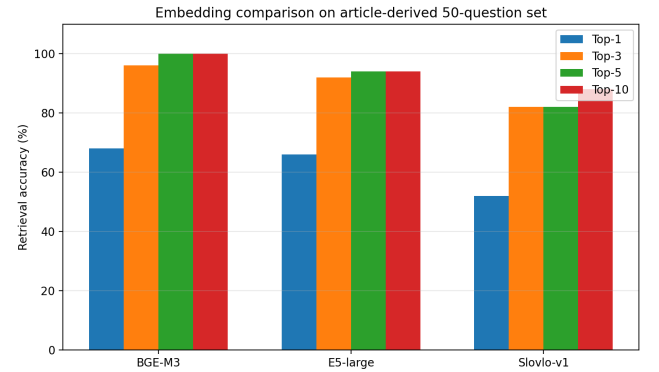


Figure 2. Embedding comparison on the final dataset.

We tried several retrieval improvements. Query expansion on the 30 question set improved Top-1 from 66.67% to 76.67% and Top-3 from 86.67% to 93.33%, but slightly low-

ered Top-5, so we treat it as promising but not final. A BGE re-ranker did not reliably improve Top-1 and sometimes promoted semantically similar but legally inappropriate chunks. MMR also hurt recall: for BGE-M3, Top-3 dropped from 96% to 80% and Top-5 from 100% to 92% (Table 3). We therefore did not use re-ranking or MMR in the final dense setup.

Emb.	Mode	T1	T3	T5	T10
BGE	Dense	68%	96%	100%	100%
BGE	MMR	68%	80%	92%	100%
E5	Dense	66%	92%	94%	94%
E5	MMR	66%	70%	74%	88%
Slovlo	Dense	52%	82%	82%	88%
Slovlo	MMR	52%	64%	68%	82%

Table 3. Dense retrieval vs. MMR on the final dataset. MMR used candidate $k = 30$, final $k = 10$, $\lambda = 0.7$.

We also implemented BM25 and hybrid BM25 with dense retrieval. BM25 alone was worse than dense retrieval, but hybrid retrieval with $\alpha = 0.75$ strongly improved Top-1 from 68% to 86% on the 50 question set, while slightly reducing Top-3/Top-5 recall (Table 4). This shows that exact keyword matching helps promote correct legal articles, while dense retrieval remains better for high recall.

Method	T1	T3	T5	T10
Dense BGE-M3	68%	96%	100%	100%
BM25	54%	66%	68%	78%
Hybrid $\alpha = 0.25$	62%	76%	84%	94%
Hybrid $\alpha = 0.50$	70%	88%	94%	98%
Hybrid $\alpha = 0.75$	86%	90%	98%	100%

Table 4. BM25 and hybrid retrieval on the final 50-question set.

4.3 Generation settings

We swept four context/generation settings: 700/160, 700/220, 1200/160 and 1200/220, where the first number is context characters per chunk and the second is generated tokens. The 1200/220 setting had no cutoff endings and the fewest formatting problems, so it was used for final generation. Increasing context length does not change retrieval, it only gives the model more text from retrieved articles.

5. Final Evaluation

Table 5 shows final dense retrieval. The system achieved 100% Top-5 recall when at least one gold source was required. Single source questions were strong, but multi source questions remained difficult when all required sources had to be retrieved.

Set	T1	T3	T5	T10
All, at least one	68%	96%	100%	100%
Single source	77.5%	97.5%	100%	100%
Multi source, at least one	30%	90%	100%	100%
Multi source, all sources	0%	40%	50%	60%

Table 5. Final dense retrieval evaluation.

Hybrid $\alpha = 0.75$ improved answer correctness, correct

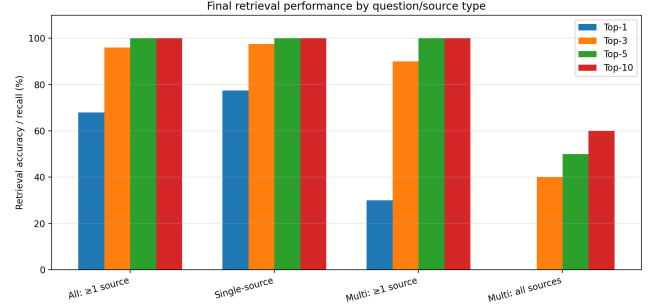


Figure 3. Final retrieval split by source type.

answers increased from 29/50 to 34/50, while partial answers decreased from 15/50 to 10/50. The number of wrong answers stayed the same. Hybrid helped single-source ranking, but still did not solve multi-source coverage.

For the dense final system, 58% of answers were correct, 30% partial, and 12% wrong (Table 6). The largest weakness was multi source reasoning with only 1/10 multi source answers being fully correct, although 7/10 were partially correct.

Type	Correct	Partial	Wrong
Easy	8/10	2/10	0/10
Hard	7/10	2/10	1/10
Comparison	3/5	2/5	0/5
Scenario	4/5	0/5	1/5
Yes/No	3/5	0/5	2/5
Tricky neg.	3/5	2/5	0/5
Multi-source	1/10	7/10	2/10

Table 6. Final answer quality by question type.

Additionally, we tested 10 general and 10 unanswerable questions, to see how the model answers when there are no sources available. The model handled many broad legal questions reasonably, but unanswerable questions were risky. It sometimes correctly refused, but sometimes hallucinated concrete facts, such as inventing a notary for a specific address. Some example inputs and outputs showing correct grounding, correct refusal and hallucination on unanswerable questions can be seen in Table 7.

Question	Observed answer behavior
Kaj pomeni služnost?	Correctly answered from SPZ: service/easement is the right to use another's thing or require omission of acts.
Kdo je danes lastnik parcele 123/4?	Correctly refused in one run: the provided legal sources do not contain current ownership data.
Kateri notar je overil pogodbo za Celovško 55?	Failure case: hallucinated a concrete notary, showing weak answerability detection.

Table 7. Example outputs showing correct grounding, correct refusal, and hallucination on unanswerable questions.

6. Discussion

RAG substantially improved grounding compared with no-RAG generation, across GaMS-2B, GaMS-9B, and GaMS3-12B, wrong answers decreased from 26/90 to 7/90, while

correct answers increased from 23/90 to 63/90. However, high Top-5 retrieval recall is not sufficient because the generator only sees the selected context chunks. With `context_n=2`, if the gold article is ranked third or lower, the model cannot use it and may answer from a related article, rely on parametric knowledge or guess. For single source questions this worked reasonably because the correct or related article was often in the top two. For multi source questions it was not enough.

The context experiment confirmed, that increasing `context_n` to 5 improved evidence visibility for multi source questions, but also increased verbosity and citation risk. This suggests that a fixed context size is suboptimal. Simple questions should use few chunks, while multi source, comparison or scenario questions should use more.

The hybrid BM25 with dense experiment showed that keyword matching is valuable for legal retrieval. Hybrid $\alpha = 0.75$ improved Top-1 and fully correct answers, but dense BGE-M3 still had the best broad recall.

7. Future Work

The most direct improvement is adaptive retrieval. Use `context_n=2` for simple questions and more chunks for multi source or comparison questions. Another improvement is query decomposition, where a complex legal question is split into sub queries and evidence is retrieved separately for each part. This is especially relevant for multi source questions.

We also plan to test query expansion on the final 50 question set to evaluate hybrid retrieval more thoroughly and compare against additional models such as Qwen2.5-7B-Instruct. Other future work includes BM25 with dense fusion with better calibration, ColBERT-style late interaction retrieval, token based chunking with overlap and conversational RAG where follow up questions are rewritten into standalone questions before retrieval. If local RAG context is insufficient, future systems could fall back to trusted official sources such as PISRS or court/authority websites, for legal QA. Unrestricted web search would be unsafe and could provide false information.

8. Conclusion

We built a complete Slovenian legal RAG system over PISRS legal documents. The final dense baseline achieved 100% Top-5 retrieval recall on the 50 question article derived dataset, with 58% fully correct answers and 30% partially correct answers. Hybrid BM25 with dense retrieval improved Top-1 from 68% to 86% and fully correct answers from 29/50 to 34/50, although multi source questions remained difficult. The main lesson is that legal RAG quality depends more on source selection and evidence coverage than on model size alone.

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